

Official Data & Metadata Reference

Youth Development & Economic Opportunity Analytics

Purpose	A concise technical reference for the four source indicators used in Youth Development & Economic Opportunity Analytics. It records definitions, units, methodological considerations, source-data checks, and official documentation.
Sources	World Bank and International Labour Organization(ILO/ILOSTAT)

How this reference is used

The project follows a documentation-first workflow: understand the official indicator definition and methodology, inspect the downloaded data, clean only what is necessary, and then analyze.

Official source terminology is retained throughout the project. Metadata and observation-status fields should be checked before they are removed or transformed.

1. Indicator Reference

Indicator	Source	Unit	Key methodological point
GDP per capita, PPP (constant 2021 international \$)	World Bank	International \$	PPP-adjusted GDP per person, expressed in constant 2021 international dollars. Designed for comparison across countries and over time.
Lower secondary completion rate, total (% of relevant age group)	World Bank	% of relevant age group	Defined by the World Bank as the gross intake ratio to the last grade of lower secondary education. Values can exceed 100% because of age differences, repetition, and denominator effects.
Unemployment, youth total (% of total labour force ages 15–24) — ILO modelled estimate	ILO / ILOSTAT	%	ILO modelled series can combine reported observations with statistically imputed estimates to create more complete and internationally comparable series.
Share of youth not in employment, education or training (NEET)	ILO / ILOSTAT	%	Measures youth who are neither employed nor participating in education or training. It is broader than unemployment alone.

2. Interpretation Rules

GDP per capita: This is GDP per capita, not total GDP. PPP adjusts for differences in price levels between countries, while constant 2021 dollars help make values comparable over time.

Education: Do not automatically treat values above 100% as errors. The official metadata explains why this can occur.

ILO modelled estimates: Where a suitable directly reported observation is unavailable, ILO statistical models can estimate a value using available information. Such an estimate should not be described as a direct survey observation.

NEET sex dimension: The raw NEET dataset contains Total, Male, Female, and other sex categories. For the project's country-year analytical dataset, the Total category was retained.

NEET source and status: NEET_Source records the source of an observation, while NEET_Status records an observation-status flag such as Break in series or Unreliable. These are metadata fields, not additional measurements.

Observation-status fields: Do not delete ILO status/metadata columns before checking the official documentation. They may indicate how an observation was produced or classified.

3. Project Data Workflow

Step	Stage	Project rule
1	Official documentation first	Understand the indicator, unit, definition, and methodology.
2	Inspect downloaded data	Check fields, types, coverage, missingness, duplicates, and metadata.
3	Clean only what is necessary	Standardize formats without removing valid observations simply because they look unusual.
4	Align the indicators	Use country-year keys to build a comparable analytical dataset.
5	Analyze	Perform descriptive analysis, association analysis, and visualization with documented limitations.

4. Recorded Source-Data Checks

These checks describe the downloaded source files and should not be confused with the final merged analytical dataset.

Dataset	Recorded size	Years	Coverage	Check / note
GDP	8,430 rows × 41 columns	1990–2024	246 countries	0 missing OBS_VALUE; 0 duplicate country-year rows
Education	5,939 rows × 41 columns	1970–2025	242 countries	0 missing OBS_VALUE; 0 duplicate country-year rows
Youth unemployment	265 rows × 70 columns	Source file	Source file	Downloaded CSV contained four metadata/header rows; this was the cleaned wide source table at that stage.
NEET	13,748 rows × 9 columns	1970–2027	278 countries/areas	14 missing obs_value in the raw dataset; 0 duplicate country-year-sex rows. Sex categories include Total, Male, Female, and Other. After retaining Total,

				duplicate country-year check = 0.
--	--	--	--	--------------------------------------

The final project dataset and its country-year intersection are documented in the analysis notebook and README.

5. Official Documentation

Consult these official sources whenever a methodological decision needs verification.

World Bank — GDP per capita, PPP metadata

https://data360files.worldbank.org/data360-data/metadata/WB_WDI/WB_WDI_NY_GDP_PCAP_PP_KD.pdf

World Bank — Lower secondary completion rate metadata

<https://databank.worldbank.org/metadataglossary/world-development-indicators/series/SE.SEC.CMPT.LO.FE.ZS>

ILOSTAT — ILO modelled estimates / WESO methodology

<https://ilostat.ilo.org/dataviz/weso/>

ILOSTAT — Data and labour statistics

<https://ilostat.ilo.org/data/>

ILOSTAT — Methods and standards

<https://ilostat.ilo.org/methods/>

6. Reference Principles

Valid observations should not be removed solely because they are visually extreme. Modelled estimates should not be represented as directly observed survey values. Source metadata should remain distinguishable from the project's analytical findings.